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OFFICIAL STANDARD OPERATIONAL PROCEDURES

EduPath Platform System Manual

A comprehensive guide detailing platform features, workflows, and best practices across all four stakeholder roles: **Student**, **Teacher**, **Parent**, and **Administrator**.

Version: **3.0 Production** Date: **October 2026** Platform: **EduPath AI Learning Engine** Audience: **All Stakeholders**

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Covers Profile & Pricing Setup, Question Bank Management, Test Generation, Real-Time Cheating Detection Heuristics, Live Classroom Tutoring, and Earnings Ledgers.

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Covers Student Account Linkage, Weekly/Monthly AI Progress Digests, Subject Mastery Heatmaps, Activity Feeds, and Direct Teacher Collaboration.

🛡️ Part 4: Administrator Manual
Covers User Moderation, Academic Registries (Subjects & Grades), Question Approval, Theme & Branding Customization, and DB Mode Configuration (Mock vs. PostgreSQL).

👨‍🎓 Part 1: Student (User) Operational Manual

ROLE: STUDENT

Core workflows for learning, booking instructors, attending live classes, and tracking performance.

1.1 Account Setup & Diagnostic Onboarding

Getting started on EduPath begins with account creation and baseline calibration:

1. **Registration (`/register`)**
   Select **Student** role, choose your registered Grade Level (Grades 4–12), provide your name and email, and set a password.

2. **Diagnostic Placement Quick-Check (`/test-enhanced/test-005`)**
   Immediately take or schedule the 5-minute adaptive placement assessment. The AI engine evaluates foundational grasp to configure your personalized dashboard recommendations.

3. **Parent Account Linkage**
   Share your registered student email with your parents so they can pair accounts and receive weekly progress digests.

1.2 Selecting Teachers, Booking & Payments

ACTION	LOCATION	PROCESS & DESCRIPTION
Browse Directory	`/teachers`	Filter instructors by subject area (Mathematics, Science, English, etc.), view verified ratings (4.9 ★), and compare rates ($35/session).
Schedule Session	Modal on Teacher Card	Click **"Book Lesson"** to open the Booking Modal. Choose a custom lesson focus, date, and 45-minute time slot.
Payment Tiers	Checkout Step 2	• **Single Lesson:** $35.00 one-time. • **Monthly Study Pass:** $119.00 (4 lessons, save 15%). • **Study Pass Credit:** $0.00 (auto-deducts 1 remaining credit).
Confirmation & Calendar	Confirmation Step 3	Receive an instant receipt ID and click **"Add to Calendar (.ics)"** to sync with Apple, Google, or Outlook calendars.

1.3 Attending Live Lessons (`/classroom/:lessonId`)

📹 Live WebRTC Stage

Simulated 1080p stream with teacher video, student PiP camera, mic controls, screen share, and raise-hand notifications.

🎨 Digital Whiteboard

Real-time collaborative canvas with colored markers, formula demonstrations, and eraser tools for step-by-step problem solving.

📝 In-Session Notes Pad

Private scratchpad auto-saves student notes locally with a 1-click **Export .txt** download button.

> 💡 **Post-Lesson Assessment Bridge:** At the conclusion of each live session, click **"Finish & Launch Checkpoint"** to take a targeted 5-question test that validates concept retention and updates your mastery graph.

👨‍🏫 Part 2: Teacher Operational Manual

Curriculum authoring, question banks, cheating surveillance, live tutoring, and gradebooks.

ROLE: TEACHER

2.1 Teacher Dashboard Hub (`/teacher-dashboard`)

The Teacher Dashboard provides a unified command center organized into distinct tabs:

DASHBOARD TAB	CAPABILITIES	KEY ACTIONS
Overview	Active student enrollments, upcoming scheduled bookings, monthly revenue charts, and recent earnings ledger.	Review weekly lesson load, launch scheduled classroom rooms, track student retention.
Question Bank	Search, filter, and author MCQ and True/False questions by subject, grade level, difficulty, and weight.	Click **"Add Question"**, configure explanation text, or trigger AI Question Generation.
Test Builder	Assemble assessment templates from question banks or generate adaptive multi-tier test configurations.	Set test duration, passing thresholds, adaptive difficulty toggles, and publish quizzes.
Curriculum Units	Organize subjects into units and lessons; upload video links and downloadable PDF materials.	Attach slides and study materials accessed by enrolled students in their syllabus feed.

2.2 AI Cheating Detection & Integrity (`/admin/cheating-detection`)

EduPath employs non-invasive heuristic tracking during student assessment sessions:

🔍 Monitored Triggers

- **Tab-Switching:** Records window blur events when students switch screens.
- **Pacing Anomalies:** Flags answers submitted in <3 seconds on hard items.
- **Score Jumps:** Identifies uncharacteristic performance spikes.

⚖️ Educator Actions

Instructors can review flagged sessions, view the confidence score (0–100%), inspect timeline logs, and mark flags as **"Reviewed"**, **"Cleared"**, or **"Flagged for Retest"**.

2.3 Conducting Live Tutoring & Managing Payouts

1 **Entering Room**

Launch the scheduled session from your Overview calendar (`/classroom/:bookingId`). Confirm camera and mic permissions.

2 **Guiding the Agenda**

Check off learning goals in real-time on the right sidebar agenda. Use the shared whiteboard to demonstrate equations.

3 **Session Payout Ledger**

All completed tutoring sessions automatically credit your teacher balance. View your monthly earnings distribution and ledger under Dashboard > Overview > Ledger.

👨‍👩‍👧 Part 3: Parent Operational Manual

Student monitoring, AI-generated summary digests, progress milestones, and alerts.

ROLE: PARENT

3.1 Parent Dashboard Overview (`/parent-dashboard`)

The Parent Dashboard gives guardians complete transparency over their children's academic journey without technical jargon:

🔄 Child Switcher
Easily switch between linked students (e.g., Alex Johnson Grade 8, Jordan Johnson Grade 6) to view individual statistics.

📈 Mastery Heatmaps
Visual progress bars across Mathematics, Science, English, and History showing current competency percentages.

⚡ Live Activity Stream
Real-time log of tests started, completed quizzes, streak milestones, and live tutoring sessions attended.

3.2 AI Parent Report Summarizer (`/ai/parent-reports`)

EduPath translates complex psychometric quiz telemetry into natural-language parent summaries:

REPORT SECTION	WHAT IT EXPLAINS	PARENT BENEFIT
Executive Summary	A 3-sentence plain-English breakdown of the student's weekly effort and achievements.	Quick grasp of overall trajectory without reading raw test sheets.
Key Highlights	Bullet points celebrating mastered subjects, study streaks, and high accuracy areas.	Immediate positive reinforcement points for parents to praise.
Areas of Concern	Specific conceptual gaps (e.g., *"Word problems involving two-step inequalities"*).	Identifies exactly where the student needs guidance.
Actionable Next Steps	Concrete recommendations (e.g., *"Schedule 1 tutoring session with Teacher Demo"*).	Eliminates guesswork about how to support the child at home.

📥 **Export & Share:** Parents can click **"Export as PDF"** to print official report cards or **"Email Report"** to forward copies to school counselors.

🛡️ Part 4: Administrator Operational Manual

System architecture, user roles, content versions, registries, themes, and database operations.

ROLE: ADMIN

4.1 Admin Command Center & Management Routes

MANAGEMENT MODULE	ROUTE	ADMINISTRATIVE FUNCTION
Admin Dashboard	`/admin-dashboard`	System KPIs: Total users, question count, completed tests, and server health.
User Accounts & Roles	`/admin/users`	Activate/deactivate accounts, change user roles (Student ↔ Teacher ↔ Admin), edit grade levels.
Question Bank Admin	`/admin/questions`	Global question repository, bulk status reviews, answer explanation verification.

MANAGEMENT MODULE	ROUTE	ADMINISTRATIVE FUNCTION
Bulk Question Import	`/admin/bulk-import`	Import hundreds of questions via standardized JSON/CSV templates with syntax validation.
Academic Registries	`/admin/platform-settings`	Add or remove platform-wide Subjects and Grade Levels without code modifications.
Content Customization	`/admin/customization`	Configure platform branding, hero titles, diagnostic CTA toggles, and color palettes.
Database Sync & Drivers	`/admin/system`	Toggle active database mode (Persistent JSON vs. PostgreSQL), run SQL seed transactions.

4.2 System Operations & Database Migration Guide

1 **Database Modes**

EduPath includes dual-engine persistence: **Persistent Local JSON** (zero-dependency local development and mock resilience) and **PostgreSQL** (enterprise production).

2 **Switching Modes**

Admins can toggle the driver in `/admin/system` or configure `DATABASE_URL` in `.env`. The client auto-hydrates state seamlessly via `hydrateDatabaseState()`.